



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
19-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

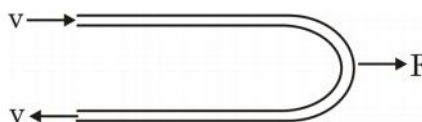
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A horizontal tube of uniform cross-sectional area A is bent in the form of U as shown in figure. If the liquid of density ρ enters and leaves the tube with velocity v , then the external force F required to hold the bend stationary is :-

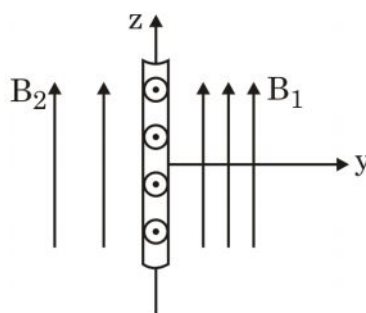


- (A) $F = 0$
- (B) $\rho A v^2$
- (C) $2\rho A v^2$
- (D) None of these
2. The activity of a radioactive sample goes down to about 6% in a time of 2 hours. The half life of sample in minutes is about :-
- (A) 30
- (B) 15
- (C) 60
- (D) 120

3. A planet of mass m is revolving in an elliptical orbit about the sun with an orbital period T . If M_{sun} be the mass of sun and A be the area of orbit, then the angular momentum of the planet is :-

- (A) $\frac{mA}{2T}$
 (B) $\frac{mA}{4T}$
 (C) $\frac{2mA}{T}$
 (D) $\frac{4mA}{T}$

4. Figure shows a thin metal sheet in the plane $y = 0$, for which the current of constant density flows in the positive x -direction. It is in a constant homogeneous magnetic field of value $B_0 \hat{k}$. As a result of superposition of magnetic field in region $y > 0$, the induction field $B_1 \hat{k}$ and in $y < 0$ is $B_2 \hat{k}$ where $B_1 > B_2$. Specify the correct statement :-



- (A) $B_0 = \frac{(B_1 - B_2)}{2}$
 (B) $B_0 = \frac{(B_1 + B_2)}{2}$
 (C) $B_0 = B_1 + B_2$
 (D) $B_0 = B_1 - B_2$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

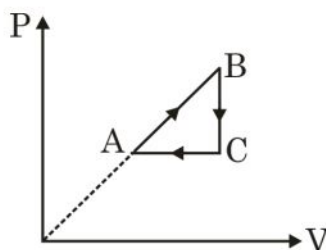
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Consider a thermodynamics cycle in a PV diagram shown in the figure performed by one mole of a monoatomic gas. The temperature at A is T_0 and volumes are related as $V_B = V_C = 2V_A$. Choose the correct options :-



- (A) The maximum temperature during the cycle is $4T_0$
- (B) Net work done by the gas during the cycle is $0.5 RT_0$
- (C) The heat capacity of the process AB is $2R$.
- (D) The efficiency of the cycle is 8.33%

6. The plates of a parallel plate capacitor with no dielectric are connected to a voltage source. Now a dielectric of dielectric constant K is inserted to fill the whole space between the plates with voltage source remaining connected to the capacitor. Select the correct statement(s) :-
- (A) The energy stored in the capacitor will become K -times the initial energy.
 - (B) The electric field inside the capacitor will decrease K -times the initial field.
 - (C) The force of attraction between the plates will become K^2 -times the initial force.
 - (D) The charge on the capacitor will become K -times the initial charge.
7. In a resonance tube experiment to determine speed of sound, air column in the pipe is made to resonate with a given fork of frequency 480 Hz. At room temperature T using a resonance tube a student finds two resonance position 16.0 cm and 51.0 cm, length of the air column for the first and the second resonance respectively. It is known that speed of the sound at 273 K is equal to 330 m/s. Then,
- (A) percentage error in the measurement of speed of sound is $\frac{4}{7}\%$
 - (B) diameter of the tube is 5 cm.
 - (C) temperature of the room T is nearly 283 K.
 - (D) percentage error in the determination of T is $\frac{8}{7}\%$

SECTION-II (i) : (Maximum Marks: 12)

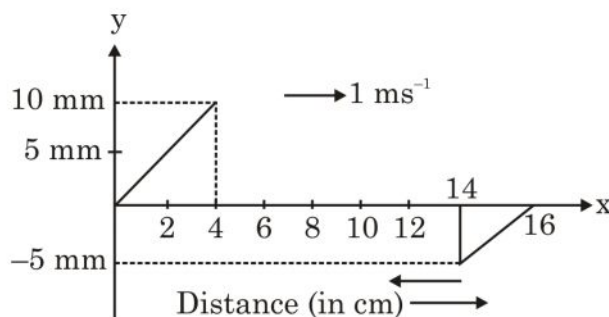
- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
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Paragraph for Questions 1 and 2

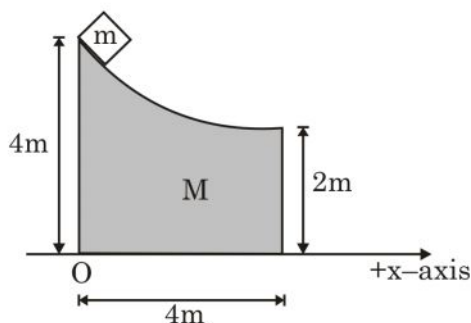
Two wave pulses are travelling in opposite direction with speed 1 ms^{-1} . Figure shows the shape of pulse at $t = 0$. Based on the above facts, answer the following questions.



1. The speed of particle (in m/s) at $x = 2 \text{ cm}$ and $t = 0$ is :-
2. The displacement of particle (in mm) at $x = 8 \text{ cm}$ and $t = 0.06 \text{ sec}$ is

Paragraph for Questions 3 and 4

A small block of mass $m = 1 \text{ kg}$ is placed over a wedge of mass $M = 4 \text{ kg}$ as shown in figure. Mass m is released from rest. All surfaces are smooth and the origin O is fixed on the ground shown. Based on above information, answer the following questions.



3. Final velocity of the wedge is (in m/s)
4. The block strikes the x-axis at (in m)

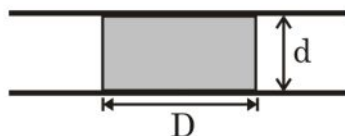
SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

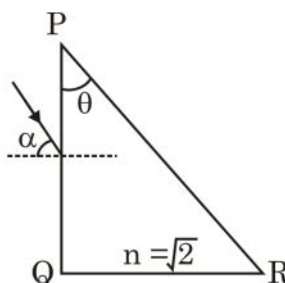
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

5. In LCR circuit current resonant frequency is 600 Hz and half power points are at 650 and 550 Hz. The quality factor is
6. In a zero gravity region, a drop of a liquid of surface tension σ assumes a cylindrical shape of diameter D between two parallel glass plates that are a distance ' d ' apart. If curved surface of the drop is at right angles to the plates as shown in the figure and force exerted by the drop on a plate is $\frac{\pi \sigma D}{n}$, then find 'n'

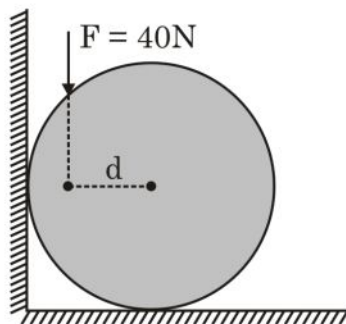


7. A parallel beam of light is incident from air at an angle ' α ' on the side PQ of a right angled triangular prism of refractive index $n = \sqrt{2}$. Light undergoes total internal reflection in the prism at the face PR when α has a minimum value of 45° . The angle ' θ ' of the prism is 'n' times 3° . Find 'n'.



8. A wire having a length L and cross-sectional area A is suspended at one of its ends from a ceiling. Density and Young's modulus of material of the wire are ρ and Y , respectively. Its strain energy due to its own weight is $\frac{\rho^2 g^2 A L^3}{pY}$. Calculate the value of p .

9. In a hypothetical system a particle of mass m and charge $-3q$ is moving around a very heavy particle having charge q . Assuming Bohr's model to be applicable to this system, the orbital velocity of mass m when it is nearest to heavy particle is $\frac{3q^2}{x\epsilon_0 h}$. Find the value of x .
10. A cylinder of radius $r = 0.1$ m and mass $M = 2$ kg is placed such that it is in contact simultaneously with a vertical and a horizontal surface as shown in figure. The coefficient of static friction is $\mu = \left(\frac{1}{3}\right)$ for both the surfaces. Find the distance d , in cm, from the centre of the cylinder at which a force $F = 40$ N should be applied so that the cylinder just starts rotating in the anticlockwise direction. Take $g = 10 \text{ ms}^{-2}$.



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

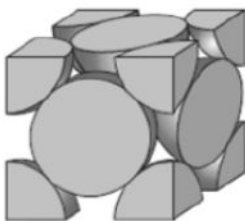
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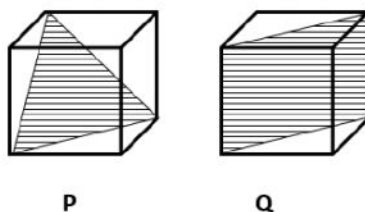
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Consider the following unit cell of a metal:



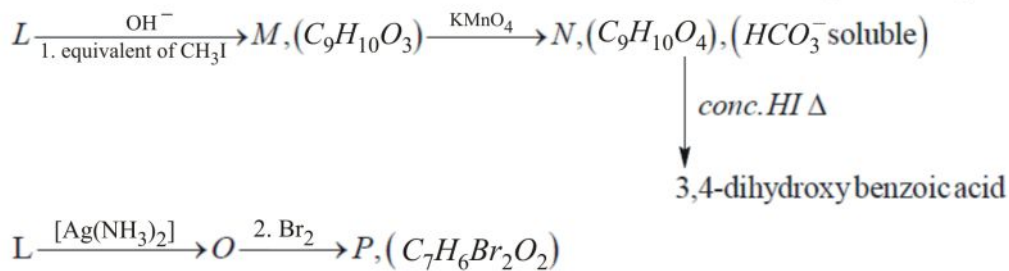
Also consider the following shaded planes P and Q representing two 2D lattices.



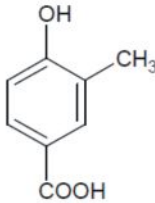
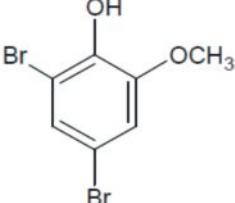
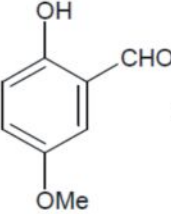
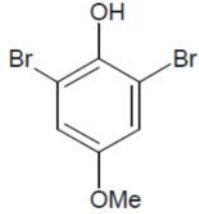
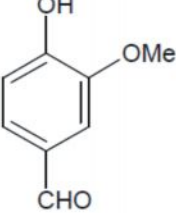
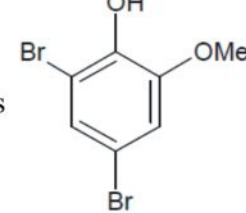
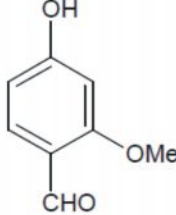
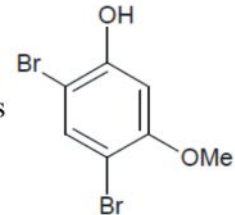
The ratio of packing fraction of P to Q is

- (A) 1.00
- (B) 1.63
- (C) 0.55
- (D) 0.74

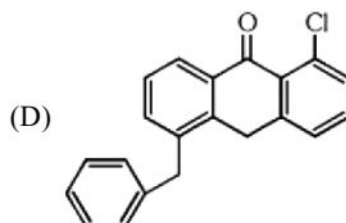
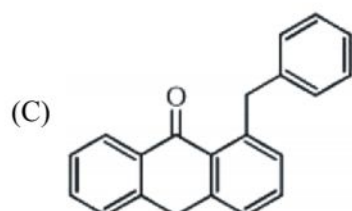
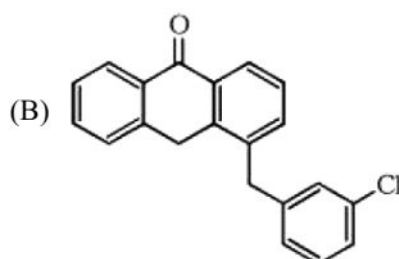
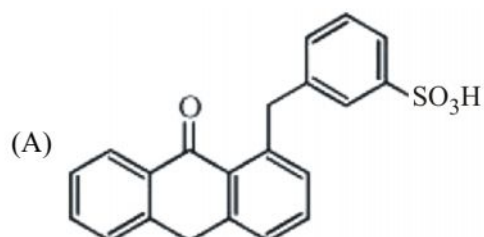
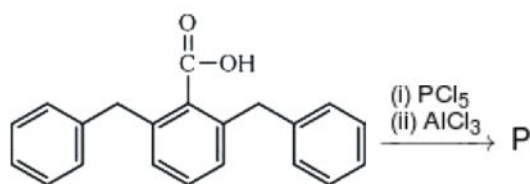
2. Vanillin (L), $C_8H_8O_3$, is isolated from vanilla beans. It give an intense colour with $FeCl_3$ and positive Tollen's test. It is not steam-distilled and does not react with HCl . It goes through following steps :



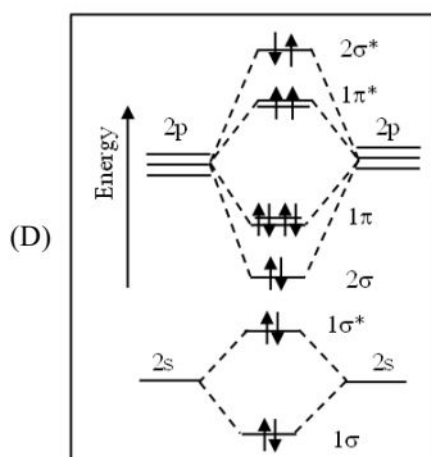
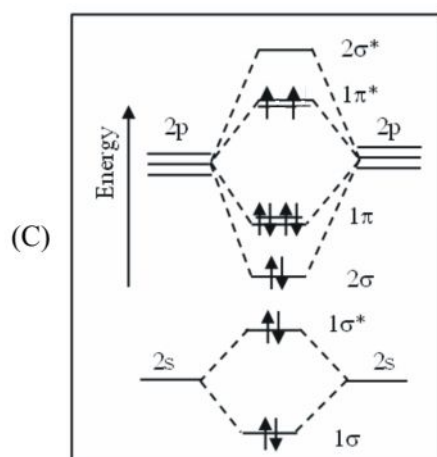
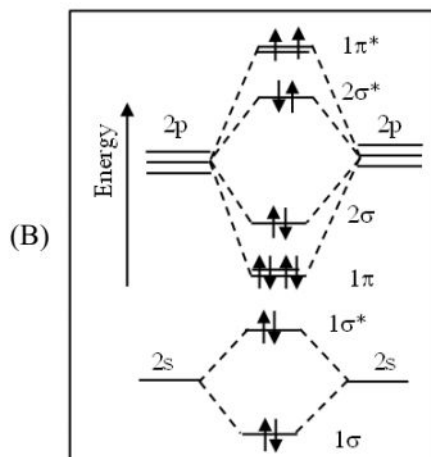
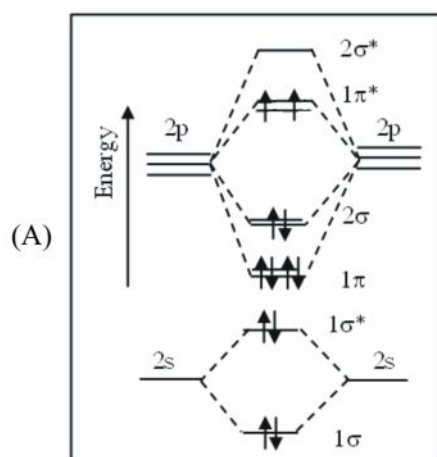
Which of the following is correct statement about various products?

- (A) L is  and P is 
- (B) L is  and P is 
- (C) L is  and P is 
- (D) L is  and P is 

3. Predict the major product (P) in the following reaction :



4. The correct molecular orbital diagram for O_2 molecule in the ground state is



SECTION-I (ii) : (Maximum Marks: 12)

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-
5. Consider one mole of a real gas A, with equation of state $P(V - b) = RT$ where 'b' is a constant. Which of the following is/are correct statement(s) about A?

(P = pressure, V=volume, T-temperature in Kelvin, $R = 8.314 \text{ J K mol}^{-1}$)

- (A) Density of A is less than ideal gas at all pressure values.
- (B) Potential energy of A, due intermolecular forces, is either greater than (or) equal to that of ideal gas
- (C) If 5J of heat is given to A, and 2J work is done by the A then, its internal energy increases by 3J, which is true for ideal gas also
- (D) Compressibility factor (Z) of A is directly proportional to 'P'.

6. When copper is reacted with dil, HNO_3 a gas is liberated. Which of the following are true for the gas?
- (A) Has 11 valence e's, where extra electron occupies an antibonding $\pi^* 2p$ orbital
 - (B) An acid anhydride
 - (C) Paramagnetic
 - (D) Neutral oxide
7. The complex(es), which can exhibit the type of isomerism shown by $[\text{Co}(\text{en})_3]\text{Cl}_3$, is(are)
[en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$]
- (A) $[\text{Pt}(\text{en})(\text{SCN})_2]$
 - (B) $[\text{Zn}(\text{gly})_2]$
 - (C) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$
 - (D) $[\text{Cr}(\text{en})_2(\text{H}_2\text{O})(\text{SO}_4)]^+$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
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- For each question, enter the correct numerical value corresponding to the answer in the designated place.
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Paragraph for Questions 1 and 2

Consider the following process of folding/unfolding of a tripeptide segment in a large protein. The free energy change (ΔG) will depend on the interaction of the unfolded tripeptide with the solvent (water) and with the rest of the protein in the folded state (see below). Assume that the tripeptide is made up of one non-polar (hydrophobic, shaded) and two polar (hydrophilic; unshaded) residues. Use the following ΔG data of formation of interaction between residue with water or protein :

- a non-polar residue and the solvent (water) $\Delta G = +8 \text{ kJ mol}^{-1}$
- a non-polar residue and the rest of the protein $\Delta G = -4 \text{ kJ mol}^{-1}$
- a polar residue and the solvent (water) $\Delta G = -16 \text{ kJ mol}^{-1}$
- a polar residue and the rest of the protein $\Delta G = -14 \text{ kJ mol}^{-1}$



Tripeptide segment in unfolded form

○ polar residue

● non-polar residue

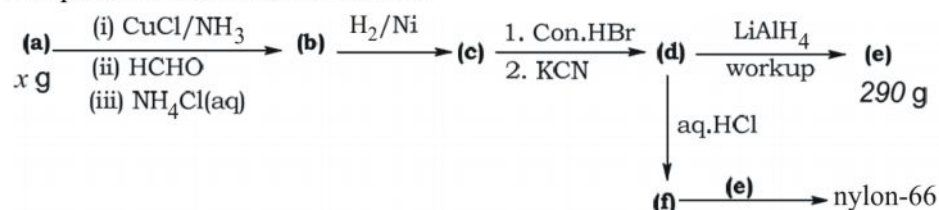


Tripeptide segment in folded form

- What is the magnitude of ΔG for the folding of the tripeptide segment in KJ/mol?
- What is the magnitude of ΔG for the folding of the tripeptide segment if all the three residues are polar in KJ/mol?

Paragraph for Questions 3 and 4

Given below is synthesis of nylon-66. Assume 100 % yield in every step and none of the organic compounds involved are in excess.



- The value of x is _____.
- The value of y is _____.

SECTION-II (ii) : (Maximum Marks: 24)

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- The answer to each question is a **NON-NEGATIVE INTEGER**
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- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

5. A solution of 0.1 M CH_3COOH is placed between parallel electrodes of cross-section area 4 cm^2 , separated by 2 cm. For this solution, resistance measured is 100Ω . The elevation in boiling point of the 0.1 M CH_3COOH solution is $x \times 10^{-5}$. The value of x is (use following information)
 $K_b = 0.5 \text{ K kg / mol}$; $\Lambda_m^\infty (\text{H}^+) = 300 \text{ S cm}^2 \text{ mole}^{-1}$; $\Lambda_m^\infty (\text{CH}_3\text{COO}^-) = 100 \text{ S cm}^2 \text{ mole}^{-1}$
6. $\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{I}_2 \xrightarrow{\text{NaOH}}$ Iodoform reaction
 How many molecules of I_2 are required for 3 molecules of CH_3COCH_3 ?
7. Find total number of inner orbital diamagnetic complexes :
 $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Cr}(\text{en})_3]^{3+}$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Cr}(\text{CN})_6]^{3-}$, $[\text{PtCl}_6]^{2-}$,
 $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Cr}(\text{CO})_6]$
8. Consider following statements :
 (1) $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightarrow 2\text{CO}(\text{g})$, it is endothermic reaction,
 (2) In copper extraction from CuFeS_2 , CaSiO_3 is main slag formed.
 (3) Brass is an alloy consisting of Zn metal.
 (4) $\text{Cu} + \text{H}_2\text{SO}_4(\text{conc.}) \rightarrow \text{CuSO}_4 + \text{H}_2$.
 (5) In Hall-Heroult process, anode is corroded forming CO_2 gas only.
 (6) Due to non-metallic impurities melting point of iron is raised.
 (7) In extraction of copper from copper pyrite, roasting process is employed.
 How many of the following are correct statements.

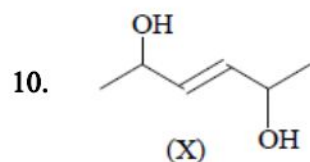
9. Consider the following precipitate/s

PbS, CuS, HgS, CdS, As₂S₃, As₂S₅, SnS, SnS₂, Sb₂S₃, Sb₂S₅

x = Total number of yellow ppt/s which soluble in yellow ammonium sulphide

y = Total number of black ppt/s which insoluble in hot & dil. HNO₃

Find the value of $\frac{x}{y}$.



Identify total number of stereoisomer's if double bond is present in cis configuration in given compound (X)

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Consider $f : [0, \infty) \rightarrow [0, \infty)$ is bijective function and it satisfies $f(x) \cdot e^{f(x)} = x$, then the value of

$$\lim_{x \rightarrow \infty} \frac{f(x)}{\ln x} \text{ is}$$

- (A) 0
(B) 1
(C) e
(D) Does not exist
2. Real number x, y, z satisfies
 $x + xy + xyz = 1$
 $y + yz + xyz = 2$
 $z + xz + xyz = 4$

If the largest possible value of xyz is $\frac{a + \sqrt{b}}{c}$, where b & c are prime numbers, then $(a + b + c)$ equals

- (A) 20
(B) 22
(C) 24
(D) 26

3. If y is implicit differentiable function of x such that $y(x+y)^2 = x$, then $\int \frac{dx}{x+3y}$ is equal to

(A) $\frac{1}{2} \ln((x+y)^2 + 1) + C$

(B) $\ln((x+y)^2 + 1) + C$

(C) $\frac{1}{2} \ln((x-y)^2 + 1) + C$

(D) $\ln((x-y)^2 + 1) + C$

Where C is constant of integration

4. Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ be a set. Let $f: A \rightarrow A$ be a function such that $f(x) \neq x \forall x \in A$, $f(x)$ is even whenever x is even and $f(x)$ is odd whenever x is odd. Then total number of onto functions satisfying these conditions will be

(A) 36

(B) 49

(C) 64

(D) 81

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable function such that $f(x) = x^3 + 5x + 1$, $g(f(x)) = x$ and $h(g(g(x))) = x \quad \forall x \in \mathbb{R}$ then
- (A) $h(0) = 7$ (B) $h'(1) = 1216$
- (C) $g'(1) = \frac{1}{5}$ (D) $h(g(2)) = 19$
6. There are 20 points lying on a circle, distance between two consecutive points is same. Three points are selected at random. X represent no. of ways to select them. Y represent no. of ways to select them such that no two adjacent objects are selected and Z represent no. of ways to select such that chosen objects are not diametrically opposite then
- (A) $X = 1140$ (B) $Y = 800$ (C) $Z = 960$ (D) $Y \cap Z = 660$
7. An ellipse having eccentricity $\frac{3}{4}$ passes through the point P (5, 4). If the nearer focus is (1,1) and equation of tangent at P on the ellipse $4x + 3y - 32 = 0$, then
- (A) Length of latus rectum of ellipse is 17.5
- (B) Length of chord through (1,1) parallel to the tangent at P is 17.5
- (C) Length of perpendicular from the focus other than (1, 1) on the tangent at P is 35
- (D) Locus of point of intersection of perpendicular tangents drawn to ellipse is $(x + 1)^2 + (y + 8)^2 = 575$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let $f(x)$ is strictly increasing and continuous function in $[0, 1]$ and for $x \in [0, 1]$ $g(x) = \int_0^x f(t) dt$, also

$\int_0^1 (f(t) + |f(t)|) dt = 6$. Local minimum value of $g(x)$ is -1 occurs at $x = a$, where $a \in (0, 1)$, then

1. $\int_0^1 g(x) f(x) dx$ is equal to
2. $\int_a^1 f(x) dx$ is equal to

Paragraph for Questions 3 and 4

A tetrahedron is formed by the planes $P_1 : y + z = 0$, $P_2 : z + x = 0$, $P_3 : x + y = 0$ and $P_4 : x + y + z = 3$.

3. The volume V of the tetrahedron is :
4. Square of distance of P_4 from the vertex opposite to it is

SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

5. Consider $f(x) = \frac{x^2 + 4}{|x|}$ and $g(x) = p|x| + 2$, $p > 0$. If the exhaustive set of values of 'p' for which $f(x) > g(x) \forall x \in \mathbb{R} - \{0\}$ is (a, b) then $a + 4b$ is
6. A fair coin is tossed 10 times and the outcomes are listed. Let E_i be the event that the i^{th} outcome is a head and E_m be the event that the list contains exactly m heads. If E_i and E_m are independent, then 'm' is equal to
7. A perpendicular is drawn from a point on the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{1}$ to the plane $x + y + z = 3$ such that the foot of the perpendicular Q also lies on the plane $x - y + z = 3$, if the coordinate of Q is (a, b, c) then $|a - b - c|$ is
8. If $\vec{a}_1 = a\hat{i} + b\hat{j}$, $\vec{a}_2 = \vec{a}_1 - (2\hat{i} + \sqrt{5}\hat{j})$ and $\vec{a}_3 = \vec{a}_1 + (2\hat{i} - \sqrt{5}\hat{j})$, then
- $$\begin{vmatrix} 1 + \vec{a}_1 \cdot \vec{a}_1 & 1 + \vec{a}_1 \cdot \vec{a}_2 & 1 + \vec{a}_1 \cdot \vec{a}_3 \\ 1 + \vec{a}_2 \cdot \vec{a}_1 & 1 + \vec{a}_2 \cdot \vec{a}_2 & 1 + \vec{a}_2 \cdot \vec{a}_3 \\ 1 + \vec{a}_3 \cdot \vec{a}_1 & 1 + \vec{a}_3 \cdot \vec{a}_2 & 1 + \vec{a}_3 \cdot \vec{a}_3 \end{vmatrix}$$
- is equal to
9. If $\sum_{r=0}^{3n} a_r(x-4)^r = \sum_{r=0}^{3n} A_r(x-5)^r$ and $a_k = 1 \forall k \geq 2n$ and $\sum_{r=0}^{3n} d_r(x-8)^r = \sum_{r=0}^{3n} B_r(x-9)^r$ and $\sum_{r=0}^{3n} d_r(x-12)^r = \sum_{r=0}^{3n} D_r(x-13)^r$ and $d_k = 1 \forall k \geq 2n$, then the value of $\frac{A_{2n} + D_{2n}}{B_{2n}}$.
10. If θ is the angle between $\vec{a}$ and $\vec{b}$ such that $\sqrt{3}(\vec{a} \cdot \vec{b}) + |\vec{b} - (\vec{a} \cdot \vec{b})\vec{a}| = 1 + \vec{b} \cdot \vec{b}$ where $|\vec{a}| = 1$ then $\operatorname{cosec}^2 \theta$ is

Space for Rough Work

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ALLEN CAREER INSTITUTE Pvt. Ltd.

Registered & Corporate Office : 'SANKALP', CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.in | Website : www.allen.ac.in

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